

Factory Reallocation & Shipping Optimization Recommendation System for Nassau Candy Distributor

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Executive Summary

Nassau Candy Distributor operates a large-scale distribution network that serves customers across multiple regions. The company currently assigns products to manufacturing factories using predefined allocation rules and legacy operational processes. While this approach has enabled business continuity, it does not account for variations in shipping performance, product profitability, or operational efficiency.

The objective of this project is to analyze historical order, sales, profit, and shipping data to identify opportunities for improving factory allocation decisions. By leveraging Python, SQL, Excel, and Power BI, this project evaluates factory performance, product-level profitability, lead times, and operational efficiency.

A rule-based recommendation engine was developed to classify products into categories such as Retain Current Factory, Monitor Performance, and Review Factory Allocation. These recommendations provide management with actionable insights that can support future optimization initiatives and improve overall supply chain efficiency.

The project ultimately transforms raw operational data into meaningful business intelligence, helping Nassau Candy Distributor make more informed and data-driven decisions regarding factory assignments and shipping operations.

Problem Statement

Nassau Candy Distributor currently assigns products to factories based on static operational rules and traditional allocation methods. As the business expands across regions and product categories, these assignment methods may no longer represent the most efficient approach.

Several operational challenges can arise from static factory allocation:

- Increased shipping lead times
- Uneven factory utilization
- Reduced operational efficiency
- Lower profitability due to logistics-related inefficiencies
- Limited visibility into the impact of alternative factory assignments

The organization lacks a structured system that can evaluate factory performance, compare product allocations, and identify opportunities for operational improvement.

This project addresses these challenges by analyzing historical sales, profit, and shipping data to evaluate factory performance and generate data-driven recommendations for future allocation decisions.

Business Objectives

Primary Objectives

The primary goal of this project is to analyze operational and sales data to identify opportunities for improving factory allocation efficiency across Nassau Candy Distributor's supply chain.

The project aims to:

1. Evaluate Factory Performance

Analyze factory-level sales, profitability, unit volume, and shipping lead times to determine overall operational effectiveness.

2. Identify High-Performing Products

Determine which products contribute the most to sales and profitability and understand how they are distributed across factories.

3. Analyze Shipping Performance

Measure lead times across products and factories to identify operational bottlenecks and opportunities for improvement.

4. Develop a Product Prioritization Framework

Create a scoring methodology that ranks products using sales, profit, and lead-time metrics.

5. Build a Recommendation System

Develop a rule-based recommendation engine that classifies products into actionable categories:

- Retain Current Factory
- Monitor Performance
- Review Factory Allocation

6. Support Data-Driven Decision Making

Provide management with actionable insights that can support future optimization strategies and improve overall supply chain performance.

Dataset Overview

The dataset contains historical operational and sales records from Nassau Candy Distributor.

Dataset Size

- Total Records: 10,194+
- Total Columns: 19
- Data Type: Structured Tabular Data

Key Fields

Field	Description
Order ID	Unique order identifier
Order Date	Date when order was placed
Ship Date	Date when order was shipped
Ship Mode	Shipping method used
Customer ID	Unique customer identifier
Country/Region	Customer location
Division	Product category
Region	Customer sales region
Product ID	Product identifier
Product Name	Product description
Sales	Revenue generated from order
Units	Quantity sold
Gross Profit	Profit earned from order
Cost	Product manufacturing cost
Lead Time	Difference between Ship Date and Order Date

Tools Used

- Python (Pandas, NumPy, Matplotlib)
- MySQL
- Microsoft Excel
- Power BI
- Tableau

Project Workflow

1. Data Cleaning and Preparation
2. SQL-Based Business Analysis
3. Exploratory Data Analysis (EDA)

4. Factory Performance Evaluation
5. Product Performance Evaluation
6. Recommendation Engine Development
7. Dashboard Development in Power BI
8. Business Recommendations and Conclusion

Data Cleaning & Preparation

Overview

Before conducting the analysis, the dataset was cleaned and transformed to ensure data quality, consistency, and readiness for analysis. Data preparation was performed using Python (Pandas) and MySQL.

Data Loading

The Nassau Candy Distributor dataset was imported into Python using the Pandas library. Initial inspection was conducted to understand the dataset structure, data types, and column distribution.

The dataset contained over 10,000 records and 19 attributes related to orders, customers, products, sales, profit, and shipping information.

Data Quality Checks

Several data quality checks were performed:

- Verified dataset dimensions
- Inspected column names
- Checked data types
- Identified missing values
- Examined duplicate records
- Reviewed numerical and categorical fields

Date Conversion

The Order Date and Ship Date columns were converted into proper datetime format to enable time-based analysis and lead-time calculations.

This conversion allowed the creation of shipping performance metrics and monthly trend analysis.

Lead Time Calculation

A new feature named Lead Time was created using the difference between Ship Date and Order Date.

Lead Time was calculated using:

Lead Time = Ship Date — Order Date

This metric was later used to evaluate shipping performance across factories, products, and regions.

Data Validation

After transformation, the dataset was validated to ensure:

- No missing values in critical fields
- Correct date formats
- Consistent numerical values
- Accurate lead-time calculations

SQL Database Preparation

The cleaned dataset was imported into MySQL for business analysis.

A dedicated database was created and SQL views were generated to support:

- Product analysis
- Regional analysis
- Factory performance evaluation
- Monthly sales analysis
- Monthly profit analysis

Outcome

The cleaned and transformed dataset served as the foundation for all subsequent exploratory analysis, factory performance evaluation, recommendation generation, and dashboard development.

SQL Analysis

Overview

After data cleaning and preparation, the dataset was imported into MySQL for business analysis. SQL was used to perform aggregations, generate insights, and create reusable views for reporting purposes.

The objective of the SQL analysis was to identify trends in sales, profitability, shipping performance, product demand, and regional performance.

Database Creation

A dedicated MySQL database was created to store the cleaned Nassau Candy Distributor dataset. The dataset was imported into a structured table for analysis and reporting.

SQL-Based Business Analysis

Multiple SQL queries and views were developed to answer key business questions.

1. Division Performance Analysis

The analysis evaluated the performance of each product division based on total sales and gross profit.

Key metrics analyzed:

- Total Sales by Division
- Total Gross Profit by Division
- Division Contribution to Overall Revenue

This analysis helped identify the most profitable business segments.

2. Regional Performance Analysis

Regional analysis was conducted to understand sales and profitability across different geographical regions.

Key metrics analyzed:

- Total Sales by Region
- Gross Profit by Region
- Unit Sales by Region

The analysis highlighted the strongest and weakest performing regions.

3. Ship Mode Analysis

Shipping methods were evaluated to understand operational performance.

Key metrics analyzed:

- Sales by Ship Mode
- Profit by Ship Mode
- Average Lead Time by Ship Mode

This analysis provided visibility into the effectiveness of different shipping methods.

4. Product Performance Analysis

Product-level analysis was performed to identify high-performing and low-performing products.

Key metrics analyzed:

- Product Sales
- Product Profitability
- Units Sold

This analysis later supported the recommendation engine.

5. Monthly Sales Trend Analysis

Monthly sales trends were analyzed using Order Date information.

Key metrics analyzed:

- Monthly Revenue
- Sales Growth Trends
- Seasonal Variations

The results helped identify periods of strong and weak business performance.

6. Monthly Profit Trend Analysis

Monthly profit analysis was performed to evaluate profitability patterns over time.

Key metrics analyzed:

- Monthly Gross Profit
- Profit Growth Trends
- Profitability Fluctuations

This analysis provided insights into the financial performance of the business.

SQL Views Created

Several SQL views were created to simplify reporting and dashboard development:

- Top Division View
- Region Performance View
- Ship Mode Analysis View
- Product Performance View
- Monthly Sales View
- Monthly Profit View

These views enabled efficient querying and reusable reporting structures.

Key Outcome

SQL analysis provided a strong foundation for understanding operational performance and identifying business trends. The insights generated through SQL were later used in Python-based exploratory analysis and Power BI dashboard development.

Exploratory Data Analysis (EDA)

Overview

Exploratory Data Analysis (EDA) was conducted using Python libraries including Pandas, NumPy, and Matplotlib. The objective was to understand the structure of the dataset, identify patterns, evaluate business performance, and uncover insights that could support factory optimization decisions.

EDA played a critical role in understanding relationships between sales, profitability, shipping performance, and product demand.

Sales Analysis

Sales analysis was performed to understand revenue distribution across products, divisions, factories, and regions.

Key observations included:

- A small group of products contributed the majority of total revenue.
- Chocolate products generated significantly higher sales compared to Sugar and Other product categories.
- Sales performance varied across regions and factories.

The analysis helped identify revenue-driving products and business segments.

Profit Analysis

Gross Profit was analyzed to evaluate the financial performance of products and factories.

Key observations included:

- High-selling products generally generated strong profitability.
- Certain products generated relatively low profit despite contributing to overall sales volume.
- Profitability varied significantly across factories.

Profit analysis helped identify products and factories with the greatest financial contribution.

Lead Time Analysis

Lead Time was calculated as the difference between Ship Date and Order Date.

Key observations included:

- Lead Time varied across products and factories.
- Certain products experienced noticeably higher average lead times.
- Lead Time became an important factor in evaluating factory performance and recommendation logic.

This metric served as a key operational performance indicator throughout the project.

Product Analysis

Product-level analysis was conducted to identify top-performing and low-performing products.

Key metrics evaluated:

- Sales
- Gross Profit
- Units Sold
- Average Lead Time

The analysis revealed that a small number of Wonka-branded products generated the majority of sales and profit.

These products later became priority candidates for factory allocation evaluation.

Factory Analysis

Factory-level exploration was conducted to compare operational performance.

Metrics analyzed included:

- Total Sales
- Gross Profit
- Units Sold
- Average Lead Time

The analysis highlighted differences in factory productivity and operational effectiveness.

This formed the basis for the Factory Efficiency Score developed later in the project.

Regional Analysis

Regional performance was evaluated to understand geographical demand patterns.

Metrics analyzed included:

- Regional Sales
- Regional Profit

- Unit Volume
- Shipping Performance

The analysis identified strong-performing regions as well as regions with lower operational efficiency.

Time Series Analysis

Monthly sales and profit trends were analyzed using Order Date information.

Key objectives included:

- Understanding business growth patterns
- Identifying seasonal fluctuations
- Monitoring monthly revenue performance

Time-based analysis provided valuable insights into business performance over time.

Visualizations Created

Several visualizations were developed to support data exploration:

- Monthly Sales Trend
- Monthly Profit Trend
- Product Performance Charts
- Factory Performance Charts
- Regional Performance Charts
- Lead Time Analysis Charts

These visualizations improved understanding of operational and financial performance.

Outcome

Exploratory Data Analysis transformed raw data into meaningful business insights and provided the analytical foundation for factory performance evaluation, product prioritization, and recommendation generation.

Factory Performance Analysis

Overview

A comprehensive factory-level performance analysis was conducted to evaluate operational efficiency across Nassau Candy Distributor's manufacturing network.

The analysis focused on four key performance indicators:

- Total Sales

- Gross Profit
- Units Sold
- Average Lead Time

The objective was to identify high-performing factories, underutilized facilities, and opportunities for operational improvement.

Factory Performance Summary

Five manufacturing facilities were evaluated:

- Lot's O' Nuts
- Wicked Choccy's
- Secret Factory
- The Other Factory
- Sugar Shack

Sales Performance

Factory-level sales analysis revealed significant differences in revenue contribution.

Factory	Total Sales
Lot's O' Nuts	76,340.15
Wicked Choccy's	55,352.75
Secret Factory	8,587.50
The Other Factory	1,282.25
Sugar Shack	220.98

Key Insight

Lot's O' Nuts generated the highest revenue and emerged as the strongest sales contributor within the manufacturing network.

Sugar Shack generated the lowest sales volume, indicating limited commercial contribution.

Profit Performance

Gross Profit analysis was performed to evaluate the financial effectiveness of each factory.

Factory	Gross Profit
Lot's O' Nuts	52,771.05
Wicked Choccy's	36,053.57
Secret Factory	4,344.70
The Other Factory	152.25
Sugar Shack	121.23

Key Insight

Lot's O' Nuts achieved the highest profitability, demonstrating both strong demand and efficient product performance.

Wicked Choccy's also contributed significantly to overall profitability and emerged as a high-performing manufacturing facility.

Units Sold Analysis

Unit sales were analyzed to evaluate production and demand levels.

Factory	Units Sold
Lot's O' Nuts	21,412
Wicked Choccy's	15,863
Secret Factory	884
The Other Factory	388
Sugar Shack	107

Key Insight

The majority of customer demand was concentrated within Lot's O' Nuts and Wicked Choccy's, indicating that these facilities are responsible for a substantial portion of product distribution.

Lead Time Analysis

Lead Time was evaluated as an operational performance metric.

Factory	Average Lead Time
Sugar Shack	1340.03
Secret Factory	1321.87
Lot's O' Nuts	1321.23
Wicked Choccy's	1321.08
The Other Factory	1280.28

Key Insight

Although lead-time differences were relatively small across factories, Sugar Shack recorded the highest average lead time and may require operational review.

The Other Factory demonstrated the lowest average lead time among all facilities.

Factory Efficiency Score

To support comparative evaluation, a Factory Efficiency Score was developed using ranking-based analysis.

The score incorporated:

- Sales Rank
- Profit Rank
- Lead Time Rank

Lower scores indicate stronger overall performance.

Factory Efficiency Ranking

Rank	Factory	Efficiency Score
1	Lot's O' Nuts	5
2	Wicked Choccy's	6
3	The Other Factory	9
4	Secret Factory	10
5	Sugar Shack	15

Findings

High-Performing Factories

Lot's O' Nuts and Wicked Choccy's demonstrated strong commercial and operational performance.

These factories generated:

- Highest sales
- Highest profit
- Highest unit volume
- Competitive lead times

Improvement Candidates

Sugar Shack recorded the lowest overall efficiency score due to:

- Lowest sales contribution
- Lowest profitability
- Highest average lead time

This facility was identified as a candidate for further operational review.

Outcome

Factory-level analysis provided a clear understanding of manufacturing performance and established the foundation for product prioritization and recommendation generation.

The insights generated in this stage were later incorporated into the Factory Optimization Recommendation System.

Product Performance Analysis

Overview

Product-level analysis was conducted to identify the products that contributed the most to sales, profitability, and operational performance.

The objective was to determine which products should remain in their current factory assignments and which products may require further operational review.

The following metrics were evaluated:

- Total Sales

- Gross Profit
- Units Sold
- Average Lead Time
- Product Priority Score

Top Performing Products

The analysis revealed that a small group of products generated the majority of business revenue and profit.

Highest Revenue Generating Products

Product Name	Factory	Sales
Wonka Bar - Triple Dazzle Caramel	Wicked Choccy's	28,485.00
Wonka Bar - Scrumdiddlyumptious	Lot's O' Nuts	27,874.80
Wonka Bar - Milk Chocolate	Wicked Choccy's	26,867.75
Wonka Bar - Fudge Mallows	Lot's O' Nuts	24,890.40
Wonka Bar - Nutty Crunch Surprise	Lot's O' Nuts	23,574.95

Key Insight

These five products accounted for the majority of overall revenue and profitability within the dataset.

As a result, they were identified as strategically important products for Nassau Candy Distributor.

Profitability Analysis

The top-performing Wonka Bar products also generated the highest gross profit values.

This indicates that these products contribute significantly to both revenue generation and overall business profitability.

Key Insight

Products that consistently perform well across sales and profit metrics should remain a priority focus for factory allocation and operational planning.

Lead Time Analysis

Lead Time was analyzed at the product level to identify potential operational concerns.

Products with Higher Lead Time

Product Name	Factory	Lead Time
Hair Toffee	The Other Factory	1455.25
Everlasting Gobstopper	Secret Factory	1394.67
SweeTARTS	Sugar Shack	1383.50
Laffy Taffy	Sugar Shack	1383.10

Key Insight

These products recorded the highest average lead times and were therefore considered potential candidates for operational review.

Longer lead times may impact customer satisfaction and supply chain responsiveness.

Product Priority Score

To support decision-making, a Product Priority Score was developed using ranking-based analysis.

The score incorporated:

- Sales Rank
- Profit Rank
- Lead Time Rank

Lower scores indicate stronger overall product performance.

This methodology enabled the comparison of products using both financial and operational metrics.

Top Priority Products

Product Name	Factory	Priority Score
Wonka Bar - Scrumdiddlyumptious	Lot's O' Nuts	11
Wonka Bar - Triple Dazzle Caramel	Wicked Choccy's	12
Wonka Bar - Milk Chocolate	Wicked Choccy's	13
Wonka Bar - Fudge Mallows	Lot's O' Nuts	15

Interpretation

These products demonstrated:

- Strong sales performance
- High profitability
- Acceptable lead times

These products were considered operationally stable and suitable for retention within their current factory assignments.

Products Requiring Attention

Several products demonstrated weaker overall performance due to lower sales, lower profitability, and higher lead times.

Review Candidates

Product Name	Factory	Priority Score
Hair Toffee	The Other Factory	36
Laffy Taffy	Sugar Shack	37
SweeTARTS	Sugar Shack	38

Interpretation

These products were identified as potential candidates for factory allocation review and future optimization efforts.

Further investigation may be required to determine whether alternative manufacturing assignments could improve performance.

Outcome

Product-level analysis provided valuable insight into product profitability, operational performance, and strategic importance.

The findings from this analysis directly supported the development of the Recommendation Engine and enabled data-driven factory allocation recommendations.

Recommendation Engine

Overview

The primary objective of this project was not only to analyze historical business performance but also to provide actionable recommendations that support future operational decision-making.

To achieve this objective, a rule-based Recommendation Engine was developed using product-level performance metrics.

The Recommendation Engine evaluates products based on their sales performance, profitability, and operational efficiency and classifies them into actionable business categories.

Recommendation Methodology

A Product Priority Score was developed to rank products according to their overall business value and operational performance.

The score was calculated using:

- Sales Rank
- Profit Rank
- Lead Time Rank

The ranking-based approach enabled the comparison of products using both financial and operational indicators.

Priority Score Formula

Priority Score = Sales Rank + Profit Rank + Lead Time Rank

Lower scores indicate stronger overall performance.

Higher scores indicate weaker performance and potential optimization opportunities.

Recommendation Categories

Based on the Product Priority Score, products were classified into three categories.

Category 1: Retain Current Factory

Criteria:

Priority Score ≤ 15

Description:

Products in this category demonstrated strong sales performance, high profitability, and acceptable lead times.

These products are currently performing efficiently and do not require factory reassignment.

Business Action:

Maintain current factory allocation.

Category 2: Monitor Performance

Criteria:

Priority Score > 15 and ≤ 30

Description:

Products in this category demonstrated moderate performance levels.

Although no immediate operational changes are required, periodic monitoring is recommended to ensure continued performance stability.

Business Action:

Continue monitoring performance trends.

Category 3: Review Factory Allocation

Criteria:

Priority Score > 30

Description:

Products in this category demonstrated weaker overall performance due to a combination of lower sales, lower profitability, and higher lead times.

These products may benefit from additional operational review or alternative factory assignment strategies.

Business Action:

Evaluate potential factory reassignment opportunities.

Recommendation Results

Retain Current Factory

Products recommended for retention:

- Wonka Bar - Scrumdiddlyumptious
- Wonka Bar - Triple Dazzle Caramel
- Wonka Bar - Milk Chocolate
- Wonka Bar - Fudge MalloWS

These products consistently generated strong sales and profit while maintaining competitive operational performance.

Monitor Performance

Products requiring ongoing monitoring:

- Wonka Bar - Nutty Crunch Surprise
- Kazookles
- Wonka Gum

- Lickable Wallpaper
- Fizzy Lifting Drinks

These products demonstrated acceptable performance but should continue to be monitored for future optimization opportunities.

Review Factory Allocation

Products identified for operational review:

- Hair Toffee
- Laffy Taffy
- SweeTARTS
- Nerds
- Fun Dip
- Everlasting Gobstopper

These products recorded weaker overall performance scores and may require additional investigation to improve efficiency and profitability.

Business Value

The Recommendation Engine transforms historical business data into actionable decision-support insights.

Rather than focusing solely on reporting past performance, the system helps management identify:

- High-performing products that should remain unchanged
- Products requiring performance monitoring
- Products that may benefit from operational optimization

This approach supports data-driven decision-making and provides a scalable framework for future factory allocation analysis.

Outcome

The Recommendation Engine successfully converted analytical findings into practical business recommendations, enabling Nassau Candy Distributor to prioritize optimization efforts and improve operational visibility.

The framework can be expanded in the future through predictive analytics, machine learning models, and advanced optimization techniques.

Business Recommendations

Based on the analysis conducted throughout this project, several business recommendations are proposed to improve operational efficiency and support future factory allocation decisions.

1. Maintain Allocation of High-Performing Products

Products such as:

- Wonka Bar - Scrumdiddlyumptious
- Wonka Bar - Triple Dazzle Caramel
- Wonka Bar - Milk Chocolate
- Wonka Bar - Fudge Mallows

demonstrated strong sales performance, high profitability, and competitive lead times.

Recommendation:

Maintain their current factory assignments and continue monitoring performance to ensure operational stability.

2. Investigate Underperforming Products

Several products recorded lower sales, lower profitability, and higher lead times.

These products include:

- Hair Toffee
- Laffy Taffy
- SweeTARTS
- Nerds
- Fun Dip
- Everlasting Gobstopper

Recommendation:

Conduct additional operational reviews to identify factors contributing to weaker performance and explore opportunities for process improvement.

3. Evaluate Sugar Shack Operations

Sugar Shack recorded:

- Lowest sales contribution
- Lowest profitability
- Highest average lead time
- Lowest efficiency score

Recommendation:

Review production capacity, demand allocation, and operational processes to determine whether performance improvements can be achieved.

4. Leverage High-Performing Factories

Lot's O' Nuts and Wicked Choccy's consistently delivered strong results across sales, profit, and operational performance metrics.

Recommendation:

Use these facilities as operational benchmarks when evaluating future factory optimization initiatives.

5. Implement Continuous Performance Monitoring

Business conditions, customer demand, and operational performance can change over time.

Recommendation:

Establish periodic reviews of factory performance, product profitability, and lead-time metrics to support ongoing optimization efforts.

6. Expand Analytics Capabilities

The current recommendation framework uses rule-based scoring.

Recommendation:

Future versions of the system can incorporate predictive analytics and machine learning techniques to simulate factory reassignment scenarios and estimate business impact before implementation.

Conclusion

This project successfully analyzed operational, sales, profitability, and shipping performance data for Nassau Candy Distributor.

Using Python, MySQL, Excel, Power BI, and Tableau, a comprehensive analytical framework was developed to evaluate factory performance and product-level efficiency.

Key accomplishments included:

- Data cleaning and preparation
- SQL-based business analysis
- Exploratory Data Analysis (EDA)
- Factory performance evaluation
- Product prioritization analysis

- Development of a rule-based recommendation engine

The analysis identified high-performing factories, strategically important products, and products requiring operational review.

A Factory Efficiency Score and Product Priority Score were developed to support objective decision-making and recommendation generation.

The project successfully transformed historical operational data into actionable business intelligence, enabling more informed factory allocation and performance management decisions.

Overall, the project demonstrates how data analytics can support supply chain optimization and improve business decision-making through evidence-based recommendations.

Future Scope

While the current project provides a strong analytical foundation, several enhancements can be implemented in future versions.

Predictive Analytics

Machine learning models can be developed to predict future lead times, sales performance, and profitability across factories and products.

Factory Reallocation Simulation

A scenario simulation engine can be created to evaluate alternative factory assignments and estimate their operational impact before implementation.

Advanced Optimization Models

Mathematical optimization techniques can be used to identify optimal factory-product combinations while balancing profitability and shipping efficiency.

Real-Time Dashboards

Integration with live operational systems can enable real-time monitoring of factory performance, sales activity, and shipping metrics.

Geographic Optimization

Factory and customer location data can be combined with geographic analysis to optimize distribution routes and reduce transportation costs.

Automated Decision Support

Future versions of the recommendation engine can automatically generate optimization recommendations based on changing business conditions and operational performance.

These enhancements would further strengthen the organization's ability to make proactive, data-driven supply chain decisions.

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Tools Used:

Python | MySQL | Excel | Power BI | Tableau

Data Analyst Internship Project

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